**Case Analysis – Avoiding Misleading Proportions**

**Question 1: Truncated Y-Axis in Sales Growth**

**1. Why is the chart misleading?**

The Y-axis starts at **90 instead of 0**. Therefore, the small difference between January sales (₹95 lakhs) and June sales (₹105 lakhs) is visually exaggerated. The actual increase is only:

[
$$\frac{105-95}{95}\times100 = 10.53\%$$
]

However, the truncated axis makes the bars look as if June sales are almost twice January sales.

**2. Visualization principle violated**

The chart violates the **zero-baseline principle for bar charts**. Bar lengths should normally start from zero when representing quantities.

**3. Improved visualization method**

Use a **bar chart with the Y-axis starting at 0**. Alternatively, a **line chart** can be used because the purpose is to show monthly sales trends.

**4. Effect on business decisions**

The misleading chart may make management believe that sales have grown dramatically. This could lead to:

- Excessive inventory or production.
- Unnecessary expansion plans.
- Overestimating marketing campaign success.
- Incorrect sales forecasts and budgets.

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**Question 2: Pie Chart with Incorrect Proportions**

Total students:

[
$$420+280+200+100=1000$$
]

**1. Correct percentages**

| **Department** | **Students** | **Correct Percentage** |
|---|---|---|
| Computer Science | 420 | **42%** |
| Electronics | 280 | **28%** |
| Mechanical | 200 | **20%** |
| Civil | 100 | **10%** |
| **Total** | **1000** | **100%** |

**2. Why is the displayed chart misleading?**

The displayed slices are approximately **50%, 30%, 15%, and 15%**, which do not match the actual enrollment proportions of **42%, 28%, 20%, and 10%**.

In particular, Civil is shown as 15% instead of 10%, while Computer Science is shown as 50% instead of 42%. This gives viewers an incorrect understanding of student distribution.

**3. Alternative visualization**

A **bar chart** would be better because the enrollment numbers can be compared directly using bar lengths.

**4. Effect on institutional planning**

Incorrect proportions could lead to:

- Allocating too many classrooms to some departments.
- Hiring the wrong number of faculty.
- Misallocating laboratory resources.
- Incorrect budgeting.
- Poor planning for infrastructure and student services.

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**Question 3: Unequal Bar Widths in Revenue Comparison**

**1. Identify the graphical error**

The error is the use of **unequal bar widths** for categorical data. Bar width should remain constant when comparing different products.

**2. How do unequal widths distort perception?**

Viewers perceive the **area** of a bar, not just its height. Increasing both the height and width makes later bars appear much larger than their actual values.

Actual revenue ratio:

$$
\frac{105}{60}=1.75
$$

So Product D has **1.75 times** the revenue of Product A, not nearly three times as much.

**3. Correct visualization design**

Use:

- Equal-width bars.
- Equal spacing between bars.
- A Y-axis starting at zero.
- Revenue represented only by bar height.
- Clear labels and units: **₹ Crores**.

A simple **2D bar chart** is appropriate.

**4. Consequences for investors**

Investors may incorrectly believe that Product D is substantially more profitable or successful than Product A. This could influence:

- Investment decisions.
- Company valuation.
- Product funding.
- Portfolio allocation.
- Perception of company performance.

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**Question 4: 3D Column Chart Distortion**

**1. Why can 3D charts be misleading?**

3D charts introduce **perspective, depth, and viewing-angle effects**. These can make bars in the foreground or certain positions appear larger than they actually are.

The chart therefore adds visual effects that do not represent additional data.

**2. Perceived difference vs actual difference**

Wind:

$$500\text{ GWh}$$

Hydro:

$$550\text{ GWh}$$

Actual difference:

$$550-500=50\text{ GWh}$$

Percentage difference relative to Wind:

$$\frac{50}{500}\times100=10\%$$

So Hydro generates only **10% more energy than Wind**. If the chart makes Hydro appear 40% larger, the visual impression is significantly exaggerated.

**3. Better visualization**

Use a **simple 2D column/bar chart** with:

- Equal-width bars.
- A zero baseline.
- No 3D perspective.
- Clear numerical labels.
- Consistent scale.

**4. Importance in sustainability reporting**

Sustainability reports must communicate environmental performance accurately. Visual distortion can cause stakeholders to:

- Misjudge renewable energy performance.
- Make incorrect investment decisions.

- Set inappropriate sustainability targets.
- Lose trust in the organization's reporting.

Accurate visualization is therefore important for **transparency, accountability, and informed decision-making**.

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**Question 5: Bubble Chart with Incorrect Scaling**

**1. Why is scaling radius misleading?**

A circle's visual size is determined by its **area**, not simply its radius.

Circle area is:

$$
A=\pi r^2
$$

If the radius is directly proportional to the number of cases, then the area grows with the **square of the number of cases**.

For example:

$$
\frac{550}{150}=3.67
$$

District E has only about **3.67 times** as many cases as District A. But if radius is scaled directly to cases, its area becomes:

$$
(3.67)^2 \approx 13.44
$$

Thus, District E can appear over **13 times larger in area**, even though it has only 3.67 times the cases.

**2. Correct method for proportional bubble sizing**

The **area** of the bubble should be proportional to the number of cases.

Since:

$$
A=\pi r^2
$$

the radius should be proportional to:

[
r\propto\sqrt{\text{number of cases}}
]

This ensures that the visual area accurately represents the data.

**3. Alternative visualization**

A **bar chart** would be more suitable for comparing disease cases across five districts because the differences can be compared easily and accurately.

**4. Impact on public health resource allocation**

The misleading chart could make District E appear to have a much more severe disease problem than it actually does. Authorities might therefore:

- Allocate excessive medical resources to District E.
- Under-allocate resources to other districts.
- Misallocate healthcare workers and medicines.
- Create inaccurate disease-control priorities.

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**Summary**

| Question | Main Visualization Problem | Better Approach |
|---|---|---|
| **1** | Truncated Y-axis exaggerates differences | Zero-based bar chart / line chart |
| **2** | Incorrect pie-slice proportions | Correct pie chart / bar chart |
| **3** | Unequal bar widths exaggerate area | Equal-width 2D bars |
| **4** | 3D perspective distorts size | Simple 2D bar/column chart |
| **5** | Bubble radius incorrectly scaled | Area proportional to value / bar chart |

**Key principle:** A good visualization should make the **visual size of a data element proportional to the actual data value** and should avoid unnecessary visual effects that exaggerate or hide differences.